Comparative Analysis of Quantum and Multi-Model Deep Learning Approaches for Skin Lesion Classification

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***Abstract*—In medical image analysis, skin lesion classification is a crucial problem since early detection can greatly lower health risks. The possibility of using quantum machine learning to replace traditional deep learning models has been studied recently.This study is positioned as a comparative benchmarking analysis to empirically evaluate the practical performance of quantum-based learning models against established deep learning architectures under identical experimental conditions.In this work, a renowned research technique was used to first create a Quantum Neural Network (QNN) and a Quantum Support Vector Classifier (QSVC). Many conventional deep learning models were then employed for comparison because of simulator limitations and poor performance. To provide a fair experimental setting, EFAM-Net, MobileNetV3, ResNet18, and EfficientNet-B0 were trained and assessed using the same dataset. Classical deep learning models performed better than quantum-based models, which had a modest classification accuracy of 62–65%. ResNet18 and EFAM-Net had validation accuracy of 83.62% and 79.89%, respectively, while EfficientNet-B0 had the greatest validation accuracy of 90.21%, closely followed by MobileNetV3 with 90.15%. These findings suggest that conventional deep learning models are now more appropriate for real-world medical image categorisation since they provide more accuracy and training stability than quantum-based methods.**

***Index Terms*—Quantum neural network, QSVC, EfficientNet, MobileNetV3, ResNet18, deep learning, and skin lesion categorisation**

I. Introduction

Skin disorders represent a significant global health concern which affects individuals across all age groups. The early and accurate identification of malignant lesions is essential because their delayed detection results in increased patient suffering and death rates. In clinical practice doctors use visual assessments of dermoscopic images to diagnose skin conditions. The diagnostic process proves effective through this method yet its success relies on the dermatologist's medical background and experience because different dermatologists will achieve different diagnostic results [1]. The growing number of cases requires dependable computer-based diagnostic solutions because this method results in subjective assessments. Machine learning has become an effective method which automates skin lesion identification through its construction of automated classification systems according to research [2]. The first methods used manual image description systems which failed to detect the small color and texture changes together with lesion boundaries present in dermoscopic images according to [3] [4]. The application of deep learning through convolutional neural networks enables systems to obtain superior classification capabilities by applying core learning methods for direct image processing. The medical imaging domain benefits from transfer learning which enables ResNet [5] and EfficientNet [6] and MobileNet [7] architectures to demonstrate both high generalization performance and low processing requirements. The various experimental setups that researchers use in their studies create considerable challenges when researchers need to compare their results across multiple studies. The testing results show that the choice of preprocessing methods together with the selected dataset partitions and optimization method will have a significant effect on the resulting accuracy. The evaluation of medical image classification systems needs to assess their overall performance together with the system's performance stability.

II. Related Work

Moturi et al. [8] reported that their lung cancer detection framework performed more effectively when multiple deep learning models were used together rather than relying on a single network. By integrating predictions from different architectures, the system achieved improved accuracy and consistency in classification results. The medical image classification system benefited from ensemble methods because they improved system reliability instead of making it more difficult to understand.

Jagannadham et al. [9] conducted research on brain tumor detection by using convolutional neural networks to analyze MRI imaging data. The researchers built the system to detect visual elements in medical scans because they wanted it to learn these elements without using preexisting features. The research study proved that deeper convolutional layers could detect structural variations which existed between tumor areas and healthy tissues.

Tschandl et al. [10] introduced the HAM10000 dataset which includes dermoscopic images that they collected from various medical environments. The dataset functions as an evaluation standard to assess automated skin lesion classifiers because it showcases both actual class imbalances and genuine classification patterns.

Yamani et al. [11] studied the performance of EfficientNet models for tumor detection tasks. Their study showed that they achieved strong results through their method of fine-tuning pretrained networks on medical data which had limited training samples. The researchers showed that specialized medical fields can benefit from the precise adaptation of pretrained architectures.

Suneetha et al. [12] developed an attention-based MobileNet system which enhances disease classification accuracy while maintaining low computational needs. The system design focused on important image areas while preserving operational efficiency which made it suitable for use in medical facilities with restricted computing power.

Sankari et al. [13] studied how quantum machine learning methods can be combined with traditional deep learning methods to analyze medical images. The research showed that quantum-assisted models can complete classification tasks but their performance becomes unstable because of two factors which are simulator limitations and restricted circuit depth.

Golchha and Verma [14] developed a quantum-enhanced support vector classifier which uses quantum kernel functions to perform image classification. The method which intended to enhance nonlinear separation capabilities suffered from scalability problems because it required simulated quantum environments to operate with large medical image datasets.

Ghosh et al. [15] developed an ensemble system for skin cancer classification by merging outputs from multiple convolutional neural networks. The results demonstrated that their system provided better performance than any single model when tested with data that had imbalanced class distribution.

Although these studies contribute valuable perspectives in both classical and quantum learning approaches, they are typically evaluated in isolation. A structured experimental comparison between quantum classifiers and multiple established deep learning architectures under identical preprocessing and evaluation conditions is still limited. This observation forms the basis of the present comparative study.

III. Dataset and Preprocessing

This work used a publicly available skin lesion dataset with multiple illness classifications to evaluate the performance of both quantum-based and deep learning models. The dataset contains dermoscopic images of different skin lesions that were captured with different lighting, resolutions, and visual characteristics. Because of its diversity, the dataset is suitable for assessing real-world classification performance.

The studies in this research were conducted using the publicly available HAM10000 (Human Against Machine with 10000 training pictures) dataset, which is commonly used for benchmarking skin lesion classification systems. Dermoscopic images from a range of diagnostic categories, including benign and malignant lesions, are included in the dataset, which depicts the distribution of actual clinical data.

The distribution of samples across different lesion categories is analyzed to understand dataset balance and its impact on model training.

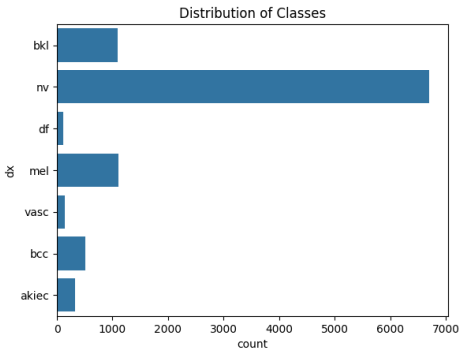

Fig. 1. Class distribution of skin lesion categories in the dataset

The chart in Figure 1 shows how the different types of skin lesions are spread out in the dataset. The dataset is highly imbalanced, with certain lesion categories containing significantly more samples than others.This kind of imbalance is common in real medical data, and if not dealt with properly, it can hurt how well the model learns. Because of this, we decided to use data augmentation techniques to make the classes more balanced and help the model perform better during training.

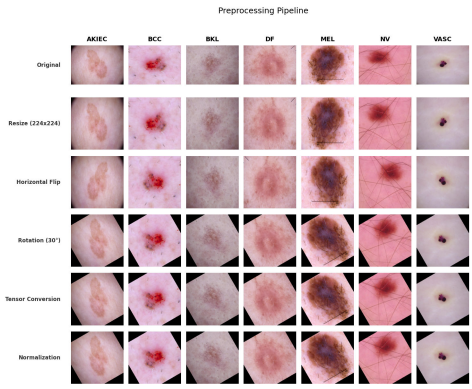

Fig. 2. Sample dermoscopic images from the skin lesion dataset

Figure 2 shows the steps taken to prepare dermoscopic images before training the model. Such images are flipped and

rotated as needed to orient them correctly before resizing them to a set resolution. issues. First of all, images are converted to tensors and further transformed to meet the input requirements for these models, which were trained earlier. Stratified splitting is one of the ways in which the splitting of the data into parts was achieved. This achieved a balance in the different classes represented throughout the various parts such that no class was overrepresented. It also made the performance results more reliable, owing to the similar mix of classes in every group. The subsets for testing, validation, and training were created using the preprocessed data. Even though the parameters of the models were learned from the training set, validation set enabled early stopping and hyper-parameter tuning. A testing set remained separated in order to ensure objective results, and it is NEVER used except in the computation of the final performance. evaluation.

IV. Proposed Methodology

The goal of the suggested approach is to evaluate and compare quantum-based learning models for skin lesion categorisation with conventional deep learning architectures.The two main stages of the entire process are multi-model deep learning optimisation and quantum model implementation.

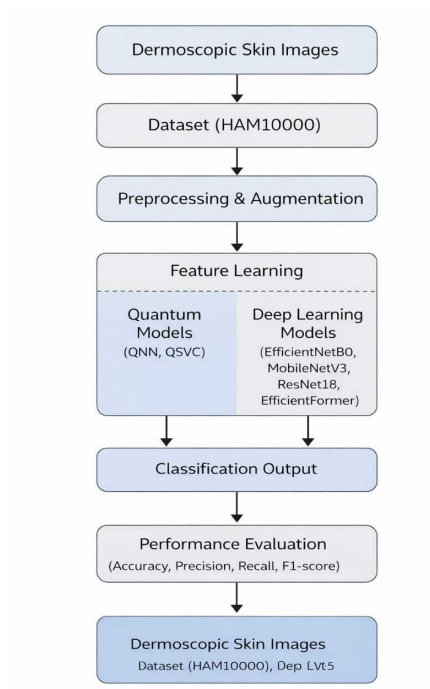

Fig. 3. Overall workflow of the proposed quantum and multi-model deep learning framework for skin lesion classification

Figure 3 illustrates the complete workflow of the proposed system. Dermoscopic skin images from the HAM10000 dataset are first subjected to preprocessing and augmentation. Feature learning is then performed using both quantum-based models (QNN and QSVC) and classical deep learning architectures. Finally, classification results are evaluated using standard performance metrics.

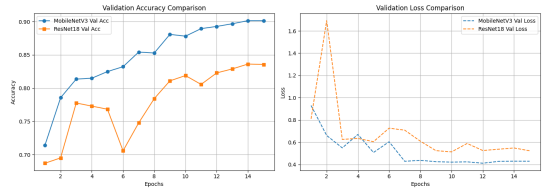

Fig. 4. Architecture overview of the deep learning models used for skin lesion classification

The architecture of the deep learning models used in this investigation, such as MobileNetV3 and ResNet18, is shown in Figure 4. The backbone of feature extraction and the classification layers modified for skin lesion prediction are shown in the figure. To increase classification performance, transfer learning was used by starting the models with pre-trained weights and then fine-tuning them on the target dataset.

To guarantee consistency and a fair comparison, both phases are carried out on the same dataset.

A. Implementation of Quantum Models

Initially, a quantum machine learning framework was used to create quantum-based classifiers on traditional simulation environments. To evaluate the viability of quantum learning for skin lesion classification, a Quantum Neural Network (QNN) and a Quantum Support Vector Classifier (QSVC) were created.Classical image features were made simpler in size and turned into quantum states using a method based on angles.

$$|\psi(x)\rangle = \bigotimes_{i=1}^{n} R_Y(x_i)\,|0\rangle$$

Where,

$$R_Y(x_i) = e^{-ix_iY/2}$$

In this process, each input value is first adjusted to a suitable range. Then, a rotation operation is applied to every qubit so that the classical data can be converted into a quantum state.The quantum models encountered real constraints such shallow circuit depth, restricted qubit availability, and simulator overhead despite their successful training.

Angle encoding was used to first reduce the dimensionality of classical picture features and then encode them into quantum states, where parameterized single-qubit rotation gates were controlled by normalized feature values. In order to overcome simulator constraints, the quantum circuits were designed with 4–6 qubits and a shallow depth of 2–3 layers.

$$U(\theta) = \prod_{l=1}^{L}\left(\prod_{i=1}^{n} R_Y(\theta_{l,i}) \cdot \mathrm{CNOT}_{i,i+1}\right)$$

where

- L represents the circuit depth,
- n denotes the number of qubits,
- θ corresponds to the trainable parameters.

A gradient-based optimizer backed by the quantum simulation framework was used to minimize a cross-entropy

loss function in order to improve the model parameters. The classical quantum simulator backend used for all quantum experiments placed restrictions on feature dimensionality and circuit depth.

$$\mathcal{L} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i)$$

All quantum experiments were conducted using a standard framework for quantum simulation. The quantum circuits were run using the state vector simulator backend for the experiments, along with a gradient-based optimizer provided within the simulation framework for parameter optimization. Due to the simulator limitations, a few qubit shallow circuits were used for the experiments to ensure convergence

B. Multi-Model Deep Learning Approach

The shortcomings of quantum-based models were addressed by a number of traditional deep learning architectures, including EfficientNetB0, EfficientFormer, MobileNetV3, and ResNet18. These models were selected because to their superior picture classification performance and appropriateness for transfer learning in medical imaging applications. Pre-trained weights were used to initialise each architecture, and the final classification layers were modified to correspond with the number of skin lesion types.

$$Z_{i,j}^{(k)} = \sum_{m} \sum_{n} W_{m,n}^{(k)} X_{i+m,\, j+n} + b^{(k)}$$

The models were fine-tuned using standardised training setups that included techniques to avoid overfitting and automatically modify learning rates to enhance performance and reliability.This study didn't use ensemble methods or dermatology-specific models to keep the comparison process simple and easy to follow. The models selected are widely used and run well even on less powerful computers, which makes it easier to compare them fairly and consistently with models that use quantum computing.

C. Learning Methodology and Feature Representation

Through the extraction of spatial, colour, and texture-based patterns from dermoscopic images, a number of convolutional layers in traditional deep learning models automatically learn feature representation. Early layers focus on low-level input such as edges and colour shifts, while deeper layers learn higher-level semantic representations relevant to lesion classification. Transfer learning is essential to this process because it speeds up convergence and reduces the requirement for extensive medical data by allowing models to repurpose information from large-scale datasets.

On the other hand, quantum-based models translate classical input into quantum states using explicit feature encoding approaches. Only low-dimensional feature vectors might be encoded by parameterised quantum circuits due to hardware and simulator limitations. Quantum models were unable to efficiently learn intricate visual patterns due to this limited representational capability. As a result, under the same experimental settings, conventional convolutional architectures performed better in feature extraction than quantum learning methods. Due to space constraints and the benchmarking nature of this study, quantum circuit structures are described conceptually rather than illustrated graphically

D. Training Configuration and Evaluation Setup

To guarantee a fair comparison,all models were trained using similar dataset splits in order to provide a fair comparison between quantum and classical learning methodologies. Classical deep learning models were optimised using adaptive gradient-based optimisers, and steady convergence was achieved by empirically adjusting learning rates. Early halting and validation-based monitoring were employed to prevent overfitting and choose the best-performing model checkpoints.

Quantum models were trained using simulator-supported optimisation techniques, with iterative parameter updates depending on measurement outcomes. Due to the stochastic nature of quantum simulations, training revealed more variance and slower convergence. Standardised metrics were used to evaluate each model's performance on an unseen test dataset in order to guarantee consistency and repeatability of outcomes across tests.

Traditionally, deep learning models use an optimizer called Adam with a learning rate equal to 0.0001 and a batch size of 32 for training. Also, a maximum of 30 epochs are employed for training these models. To prevent the model from overlearning the data it is being trained on, early stopping is used by looking at the loss function on the validation data. Additionally, a categorical cross entropy function is used for all models and data splitting for training, testing, and validation is equivalent for a fair comparison.

V. Results and Discussion

Accuracy, precision, recall, and F1-score were the standard classification measures adopted for both quantum-based and classical deep learning models. These criteria are selected to fairly assess model behaviour for several lesion categories in a medical imaging context where class imbalance and hazards from misclassification can totally change the diagnostic reliability. While accuracy denotes the overall correctness, precision and recall provide insight into false positive and false negative behavior, critical for clinical decision-making.

To account for the stochastic nature of deep learning optimisation and quantum simulations, model performance was observed to converge consistently across multiple training runs.There were small differences in how well they performed each time, but the overall order of models and how their performance changed compared to each other stayed the same.

Figure 5 displays the EfficientNetB0 training and validation accuracy and loss curves. The consistent learning behavior and optimization during the training process are demonstrated by the smooth convergence of the two curves. Good generalization with minimal overfitting can be seen from the absence

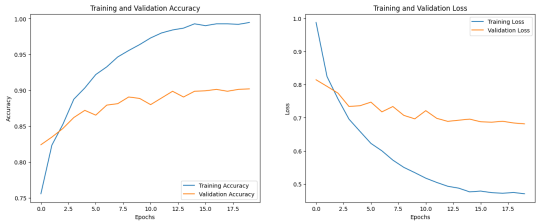

Fig. 5. Training and validation accuracy and loss curves for EfficientNetB0

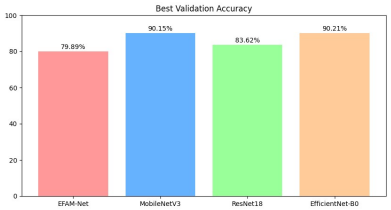

Fig. 6. Comparison of classification accuracy across different models

of distinguishable discrepancy between the two curves' performance. This is particularly important when dealing with medical picture classification problems, where it should be possible to trust the models with unknown patient data.

The quantum models were able to attain a satisfactory level of classification accuracy despite the unpredictable learning process. It was usual to discover relative differences with regard to their sensitivity to parameterization and convergence. Shallow quantum circuit depth, qubit limitations, simulator issues, and the inability of the state-of-the-art quantum simulators to efficiently encode high-dimensional picture information are some of the problems with the quantum models. As a result, the deployable scenarios for the quantum models were restricted by the ability of the models to capture complex spatial and textural details of the dermoscopic images.

On the other hand, in all the experiments, it was noticed that the training behavior of all classical deep learning models was uniform. It was noticed that over time, as the number of epochs increased, the performance also increased gradually as all the loss curves of each of the experiments were converging. In all the models analyzed, the model that showed the highest level of classification accuracy was the EfficientNetB0 and MobileNetV3. This is because it was able to learn from the small set of medical datasets.

In order to present a comprehensive comparison between the respective models, the accuracy of each of the quantum and classical models that were evaluated is represented in graphical form.

Figure 6 compares the classification accuracy achieved by the various models this study looked at. The results clearly show that traditional deep learning systems outperform quantum-based approaches under the same experimental conditions. EfficientNetB0 and MobileNetV3 demonstrated the effectiveness of modern convolutional designs on challenging medical image classification tasks, with EfficientNetB0

TABLE I
PERFORMANCE COMPARISON OF QUANTUM AND DEEP LEARNING MODELS (MACRO-AVERAGED METRICS)

Model	Acc. (%)	Precision (%)	Recall (%)	F1-score (%)
QNN	62.40	62.00	61.00	62.00
QSVC	65.10	65.00	65.00	65.00
Efam-Net	79.00	79.00	79.00	79.00
MobileNetV3	90.15	89.00	89.00	**89.00**
ResNet18	83.62	63.00	66.00	55.00
EfficientNetB0	**90.21**	64.00	57.00	60.00

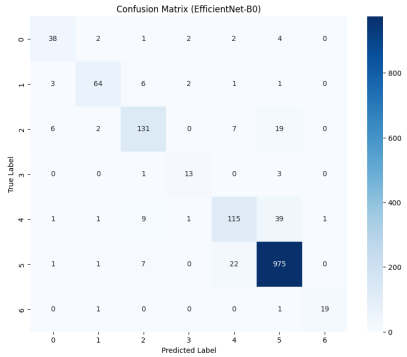

Fig. 7. Confusion matrix of the EfficientNet-B0 model on the test dataset.

providing the best results. This comparison under the same experimental conditions gives evidence of the robustness and effectiveness of state-of-the-art convolution approaches for medical image classification.

Even though accuracy was unmistakably clear from the F1-score chart as the main metric for evaluating the models, the results showed almost similar patterns of performance to validate the superiority of classical deep learning-based models to their quantum-based counterparts.

Note: Precision, recall, and F1 scores are averaged over all classes. Scores lower than the ones given result from an unbalanced dataset, rather than a misclassification.

The apparent deviation between the precision-recall-F1 metrics and the accuracy within the test set is an obvious fact of the HAM10000 dataset being imbalanced. As is well-known, accuracy always bends to the over-represented classes in general. On the other hand, F1-score is more representative of all classes, including minor classes. Therefore, a model may have high accuracy but an overall low F1-score.

Table I highlights that classical deep learning models achieve superior precision–recall balance compared to quantum-based approaches, confirming their robustness for clinical image classification.

Figure 7 illustrates the confusion matrix obtained for the EfficientNetB0 model on the test dataset.A high degree of accurate classification across the majority of skin lesion categories is indicated by the strong concentration of values along the diagonal. Lesion classes with more samples, in particular, show very high true positive rates, indicating the model's capacity to efficiently learn discriminative features.

The confusion matrix shows that classes with more training examples tend to have higher true positive rates. However, classes with fewer examples are sometimes misclassified because they are less common and can look alike other classes. This is something that often happens in real medical data, and it shows why it's important to carefully check results to make sure rare but important skin conditions aren't missed.

Due to overlapping texture and color patterns, there are a few misclassifications between visually similar lesion classes, which is to be expected in dermoscopic image analysis.

VI. Conclusion and Future Work

The present work has provided detailed comparisons about traditional deep learning approaches and quantum learning models for skin lesion classification. The main targets of this work included comparing the efficiencies of quantum machine learning approaches with traditional deep learning models in an equal state, as well as exploring the potential employment of such approaches within the field of medical image processing. When applied to complex images of skin lesions, it provides useful knowledge regarding relative merits and demerits.

Initial research attempts were made to employ quantum theory to extract discriminative attributes from images of skin lesions. Even the concept of Quantum Neural Networks and Quantum Support Vector Classifiers was valid; however, there were a number of issues that restricted their effectiveness to a certain level. For instance, the low accuracy levels and training instability were because of limitations related to quantum circuit depth being low, access to qubits being limited, reliance on the simulator being high, and processing costs being high, leading to failure of achieving the required levels of dependability for practical diagnostic systems.

However, all the metrics clearly indicated that the traditional deep learning systems performed better. EfficientNetB0, EfficientFormer, MobileNetV3, and ResNet18 are some of the systems that gave clear indications of improved classification accuracy, convergence, and generalization capabilities. Apparently, between EfficientNetB0 and MobileNetV3, the former performed better than the latter in comparison to the other model, ResNet18, which performed adequately in terms of economy and accuracy. In short, overall, it can be noted that this research work constitutes an empirical benchmarking analysis, focusing in particular on the current performance gap prevailing between quantum-based machine learning methods and already-established deep learning techniques in the context of skin lesion classification within real-world medical images.

The findings from this study further emphasize the current disconnect between the theoretical promise along with the medical image-based application possibilities for quantum machine learning. Therefore, the main focus of the future work will be to concentrate on hybrid quantum-classical models, efficient quantum feature encoding techniques, and the evaluation process. These are the practical ways of narrowing the gap between the current performance of quantum and classical methods in medical image analysis.

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